

# Mohamad Dikshie Fauzie

PRINCIPAL LOW-LATENCY INFRASTRUCTURE ARCHITECT · ELECTRONIC TRADING & DISTRIBUTED SYSTEMS

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## Summary

Principal Low-Latency Infrastructure Architect and Systems Engineer with 10+ years of mission-critical production leadership co-managing physical, network, and storage infrastructure for Japan's largest Proprietary Trading System (Japannext PTS), operating under strict Japan FSA regulatory oversight. Holds a Ph.D. from Keio University in Distributed Systems (supervised by Internet Hall of Fame founder Prof. Jun Murai and quantum pioneer Prof. Rodney Van Meter) and a B.Sc. in Physics from ITB. Proven track record executing zero-downtime exchange datacenter migrations (delivered 2 weeks early, 50% expansion headroom), 50% rack power balancing optimization, and 50% I/O throughput gains on Dell PowerScale (Isilon) for multi-petabyte market tick data capture and replay. Specializes in sub-nanosecond precision timing (NPL UK Time & Frequency metrology, IEEE 1588v2 PTP, White Rabbit, SyncE), low-latency switching (Arista 7130), kernel-bypass networking (Solarflare, NetFPGA, DPDK), and high-density colocation across Tokyo exchanges (TY3, CC2). Certified in NYIF Financial Risk Management. Trilingual: English, Japanese, Indonesian.

## Skills

|                              |                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trading Systems</b>       | Exchange Colocation (TY3, CC2), Low-Latency Switching (Arista 7130 / MetaWatch), Kernel Bypass (Solarflare Onload, EF_VI, DPDK, AF_XDP), NetFPGA, Multicast (PIM-SM/SSM), Market Data (ITCH, FAST, BBO), Order Entry (OUCH, FIX) |
| <b>Precision Timing</b>      | IEEE 1588v2 PTP (Grandmasters, Boundary/Transparent Clocks), SyncE, White Rabbit, Sub-Nanosecond Metrology, Atomic Clocks (Rubidium, OCXO), FSA / MiFID II RTS 25 Timestamp Compliance, Hardware Timestamping                    |
| <b>Systems &amp; OS</b>      | Linux Kernel Tuning (isolcpus, CPU Pinning, IRQ Affinity, HugePages, sysctl Network Stack Tuning), RHEL/CentOS, FreeBSD, High-Performance Compute (HPC) Clusters, Bare-Metal Provisioning                                        |
| <b>Storage &amp; Cloud</b>   | Dell PowerScale (Isilon) Parallel Storage (Tick Data Capture & Backtesting), LTO Tape Media Backup & Archiving, Kubernetes, Docker, OpenStack, NVIDIA AI Infrastructure (DLI Certified), AWS Cloud Fundamentals                  |
| <b>Facilities &amp; DC</b>   | Rack-Level PDU Power Telemetry, Datacenter Monitoring Dashboards, NetBox (DCIM/Rack Inventory), High-Density Thermal Balancing, Zero-Downtime Migration, Optical Metrology (DWDM, OTDR, SFPs/QSFPs), Colocation Cage Operations  |
| <b>Tools &amp; Languages</b> | Python, Bash/Shell, C/C++ familiarity, Git, NetBox, SNMP, Prometheus/Grafana, Ticket & Workload Analytics, LaTeX · English (Fluent), Japanese (Working), Indonesian (Native)                                                     |

## Work Experience

### Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

INFRASTRUCTURE LEAD (CO-MANAGER) / LOW-LATENCY SYSTEMS ARCHITECT

Oct. 2014 - Present

- Co-managed 24/7 mission-critical, ultra-low-latency datacenter and network infrastructure for Japan's primary alternative trading venue, maintaining 99.999% availability under strict Japan Financial Services Agency (FSA) compliance.
- Directed end-to-end server room migration for power-constrained trading systems; executed zero-downtime cutover 2 weeks ahead of schedule, unlocking 50% expansion capacity for matching engines and participant gateways.
- Built datacenter monitoring platforms tracking real-time PDU power/electricity draw and rack space occupancy, consolidating metrics into unified operational dashboards to achieve 50% power balancing and eliminate breaker trips.
- Conducted operational ticket data analysis and incident trend modeling to evaluate engineering workload distribution, streamline support escalation workflows, and drive data-informed capacity planning.
- Architected Dell PowerScale (Isilon) parallel storage and managed automated LTO tape media backup systems, gaining 50% I/O throughput to support multi-terabyte daily tick data logging and regulatory data archiving.
- Deployed and customized open-source DCIM platform (NetBox) for physical datacenter and rack inventory tracking, alongside optical consumables (SFPs/QSFPs, fiber), drastically accelerating fault isolation and MTTR on exchange cross-connects.
- Supervised datacenter room buildouts and low-latency cabling topologies for matching engines and broker colocation cages (Equinix TY3, ATTokyo CC2), reaching operational readiness ahead of schedule.
- Automated executive and regulatory capacity, power, and SLA compliance reporting with real-time hardware telemetry pipelines, ensuring 100% audit readiness under Japan FSA regulatory requirements.

### Keio University - JICA JST Indo Project

Tokyo, Japan

RESEARCH ASSISTANT (DISTRIBUTED SYSTEMS & NETWORKS)

Apr. 2011 - Sep. 2014

- Researched distributed network topologies, packet dissemination efficiency, and energy modeling across large-scale P2P systems and CDNs.
- Applied game theory and probabilistic gossip protocols to model message propagation, analyzing dynamic graph topologies under churn.
- Published peer-reviewed research papers in international IEEE and IEICE conferences and journals.

### Keio University

Tokyo, Japan

SYSTEM ADMINISTRATOR (CAMPUS CORE INFRASTRUCTURE)

Oct. 2007 - Sep. 2014

- Administered campus server and core backbone network infrastructure (BGP/OSPF routing, DNS, email, switches, firewalls) with high uptime.

## Certifications & Specialized Training

|             |                                                                                                                        |                |
|-------------|------------------------------------------------------------------------------------------------------------------------|----------------|
| 2022        | <b>Introduction to Time and Frequency Measurement (PTP / Clock Metrology)</b> , National Physical Laboratory (NPL), UK | Teddington, UK |
| 2020        | <b>Introduction to Metrology (Measurement Standards &amp; Uncertainty)</b> , National Physical Laboratory (NPL), UK    | Teddington, UK |
| 2010        | <b>Introduction to NetFPGA (High-Speed Line-Rate Packet Processing)</b> , Stanford University                          | Tokyo, Japan   |
| 2022        | <b>Financial Risk Management &amp; Capital Markets</b> , New York Institute of Finance / Interactive Brokers           | Online         |
| 2024        | <b>Foundations of Project Management</b> , Google Career Certificates                                                  | Online         |
| 2024        | <b>AI Infrastructure and Operations Fundamentals</b> , NVIDIA (Deep Learning Institute)                                | Online         |
| 2019        | <b>Cloud Infrastructure, Kubernetes &amp; OpenStack</b> , The Linux Foundation                                         | Online         |
| 2023        | <b>AWS Certified Cloud Practitioner</b> , Amazon Web Services (AWS)                                                    | Online         |
| 2019        | <b>Network Transformation: 5G, Core Network, Cloud &amp; NFV</b> , Intel                                               | Online         |
| 2022 - 2023 | <b>Machine Learning &amp; Mathematics for Machine Learning</b> , Stanford University / DeepLearning.AI                 | Online         |

## Publications

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Energy Consumption in Peer Assisted CDN</b>                                                                                                                               | <i>International Journal of Advanced Research in Computer Science (IJARCS) Vol.4 No.9</i><br>Aug. 2013                 |
| LEAD AUTHOR                                                                                                                                                                  |                                                                                                                        |
| <ul style="list-style-type: none"><li>Modeled and evaluated energy efficiency of peer-assisted edge CDN architectures versus traditional centralized distribution.</li></ul> |                                                                                                                        |
| <b>Assessing The Dynamics of BitTorrent Swarm Topologies Using The Peer Exchange Protocol</b>                                                                                | <i>IEICE Transactions on Communications (Special Issue on Information Network Science, Vol.E95-B, No.05)</i><br>2012   |
| LEAD AUTHOR                                                                                                                                                                  |                                                                                                                        |
| <ul style="list-style-type: none"><li>Empirically inferred dynamic overlay topologies and peer discovery mechanisms using the Peer Exchange (PEX) protocol.</li></ul>        |                                                                                                                        |
| <b>A Temporal View of The Topology of Dynamic BitTorrent Swarms</b>                                                                                                          | <i>IEEE INFOCOM Workshop on Network Science for Communication Networks (DOI: 10.1109/INFCOMW.2011.5928939)</i><br>2011 |
| LEAD AUTHOR                                                                                                                                                                  |                                                                                                                        |
| <ul style="list-style-type: none"><li>Analyzed temporal graph evolution and topological robustness of large-scale distributed swarms under dynamic churn.</li></ul>          |                                                                                                                        |

## Education

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Keio University</b>                                                                                                                                                                                                                                                                                                                          | <i>Tokyo, Japan</i><br>Graduated Oct. 2014       |
| PH.D. IN MEDIA AND GOVERNANCE (INTERNET ARCHITECTURE & DISTRIBUTED SYSTEMS)                                                                                                                                                                                                                                                                     |                                                  |
| <ul style="list-style-type: none"><li>Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).</li><li>Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer), Dr. Kenjiro Cho, Prof. Hiroshi Kusumoto.</li></ul> |                                                  |
| <b>Institut Teknologi Bandung (ITB)</b>                                                                                                                                                                                                                                                                                                         | <i>Bandung, Indonesia</i><br>Graduated Oct. 2004 |
| M.T. (MASTER OF TECHNOLOGY IN ELECTRICAL ENGINEERING)                                                                                                                                                                                                                                                                                           |                                                  |
| <ul style="list-style-type: none"><li>Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.</li></ul>                                                                                                                                                                                |                                                  |
| <b>Institut Teknologi Bandung (ITB)</b>                                                                                                                                                                                                                                                                                                         | <i>Bandung, Indonesia</i><br>Graduated Oct. 2000 |
| S.Si. (BACHELOR OF SCIENCE IN PHYSICS)                                                                                                                                                                                                                                                                                                          |                                                  |
| <ul style="list-style-type: none"><li>Researched data assimilation methods in 1D hydrodynamic fluid systems (numerical modeling, wave equations, fluid mechanics).</li></ul>                                                                                                                                                                    |                                                  |